$$\int \sec^n x\,dx = \frac{\tan x \sec^{n-2} x}{n-1} + \frac{n-2}{n-1}\int \sec^{n-2} x\,dx$$

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$\int \operatorname{cosec}^n x \, dx =$

$$-\frac{\cot x \operatorname{cosec}^{n-2} x}{n-1} + \frac{n-2}{n-1} \int \operatorname{cosec}^{n-2} x \, dx$$

Q1. $\int \cos^7 x \, dx$

Q2. $\int \cos^5 x \, dx$

3. $\int \cos^4 x \, dx$

4. $\int \cos^8 x \, dx$

For $\cot x$

$\int \cot^5 x \, dx$

$\int \cot^7 x \, dx$

For $\tan x$

$\int \tan^6 x \, dx$

$\int \tan^7 x \, dx$

For $\sec x$ and $\operatorname{cosec} x$

$\int \sec^5 x \, dx$ and $\int \operatorname{cosec}^7 x \, dx$.

Integrations:

$\int \sin x \, dx = -\cos x + c$

$\int \cos x \, dx = \sin x + c$

$\int \tan x \, dx = \ln|\sec x| + c$ or $-\ln|\cos x| + c$.

$\int \cot x \, dx = \ln|\sin x| + c$